

Algebra Solutions

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Mathematics is a game played according to certain simple rules with meaningless marks on paper.

-David Hilbert

1

Make the series of subs

$$\begin{aligned} w &= uvu^2vuv \\ &= xux^2 \\ &= xy \end{aligned} \qquad \begin{aligned} v &= u^{-1}x \\ u &= yx^{-2} \end{aligned}$$

therefore there are $|G|$ such choices since we choose x freely and must set $y = x^{-1}w$.

2

Suppose that $g \in G$. Pair all the elements up as $(a, a^{-1}g)$. By pigeonhole principle A must contain two elements in a single pair. Therefore by multiplying together we get g .

3

Let H_1 and H_2 have index a and b respectively. Then

$$|H_1 \cup H_2| = \frac{|G|}{a} + \frac{|G|}{b} - |H_1 \cap H_2| < |G|$$

since $a, b \geq 2$ and $|H_1 \cap H_2| \geq 1$. In the case of 3 subgroups one has

$$V_4 = \langle 1, 0 \rangle \cup \langle 0, 1 \rangle \cup \langle 1, 1 \rangle$$

as desired.

4

Note

$$x \circ y = (x \circ x) \circ y = (y \circ x) \circ x = ((y \circ y) \circ x) \circ x = ((y \circ x) \circ y) \circ x = (y \circ x) \circ (y \circ x) = y \circ x$$

and using commutativity

$$(x \circ y) \circ z = (z \circ y) \circ x = x \circ (y \circ z)$$

5

The multiplication rule allows us to move terms between the right and the left of an equation. IE

$$a * x = b \implies (a * x) * x = b * x \implies a = b * x$$

$$x * a = b \implies x * (x * a) = x * b \implies a = b * x$$

and the reverse implications follow by symmetry. Therefore:

$$\begin{aligned} y * x = x * y &\Leftrightarrow x * (y * x) = y \\ &\Leftrightarrow x = y * (y * x) \\ &\Leftrightarrow x = x \end{aligned}$$

The counterexample for associativity is given by the "rock paper scissors" operation:

*	a	b	c
a	a	c	b
b	c	b	a
c	b	a	c

6

Note that

$$\begin{aligned} aba &= ba^2b \\ ab &= ba^2ba^2 && \text{right multiply by } a^2 \\ ab^2 &= ba^2ba^2b \end{aligned}$$

the RHS is symmetrical and we could have just as easily derived $b^2a = ba^2ba^2b$ by performing the same steps on the opposite side. Therefore $ab^2 = b^2a$ and all the even powers of b commute with a . In particular $b^{2n} = b$ since $|b| = 2n - 1$ so b commutes with a by (b). The first relation is then

$$aba = ba^2b \implies a^2b = a^2b^2 \implies b = e$$

as desired.

7

First, we rewrite the relation $ABA^{-1}B = 1$ as $ABA^{-1} = B^{-1}$. This indicates that conjugation by A inverts B . By extension, we can move B 's past A 's using the rule:

$$B^j A^i = A^i B^{j(-1)^i}$$

Every element $C \in G$ can therefore be written in the normal form $C = A^i B^j$, where $0 \leq i \leq 3$ and $0 \leq j \leq 6$.

(a) We compute the square of an arbitrary element $C = A^i B^j \in G$:

$$C^2 = (A^i B^j)^2 = A^i B^j A^i B^j = A^{2i} B^{j(-1)^i + j}$$

We evaluate this for each possible value of $i \pmod{4}$:

$i = 0$: $C^2 = A^0 B^{j+j} = B^{2j}$. Because $\gcd(2, 7) = 1$, as j ranges from 0 to 6, the powers $2j \pmod{7}$ permute the elements. This generates all 7 elements of the subgroup $\langle B \rangle$.

$i = 1$: $C^2 = A^2 B^{-j+j} = A^2 B^0 = A^2$. This yields exactly 1 element, regardless of j .

$i = 2$: $C^2 = A^4 B^{j+j} = B^{2j}$ (since $A^4 = 1$). This yields the exact same 7 elements found in the $i = 0$ case.

$i = 3$: $C^2 = A^6 B^{-j+j} = A^2$. This yields the exact same element found in the $i = 1$ case.

Counting the distinct outcomes, there are 7 elements from the $i \in \{0, 2\}$ cases, and 1 element from the $i \in \{1, 3\}$ cases. There are 8 elements of the form C^2 in G .

- (b) Based on the casework above, the distinct squares consist of all powers of B , plus the element A^2 . Written as words in A and B , the 8 squares are:

$$1, B, B^2, B^3, B^4, B^5, B^6, A^2$$

8

We note that $a^3 = e$ since $e \in H$ and Now note that the inverse of an element has the same order as the element and we can write

$$\begin{aligned} xa &= [(xa)^{-1}]^2 = (a^2 x^{-1})^2 = a^2 x^{-1} a^2 x^{-1} \\ xa^2 &= [(xa^2)^{-1}]^2 = (ax^{-1})^2 = ax^{-1} ax^{-1} \end{aligned}$$

If we apply this identity to a segment of an expression we get:

$$\begin{aligned} x_1 a x_2 a x_3 &= x_1 a (a^2 x_2^{-1} a^2 x_2^{-1}) x_3 = (x_1 x_2^{-1}) a^2 (x_2^{-1} x_3) = x'_1 a^2 x'_2 \\ x_1 a^2 x_2 a^2 x_3 &= x_1 a^2 (a x_2^{-1} a x_2^{-1}) x_3 = (x_1 x_2^{-1}) a (x_2^{-1} x_3) = x'_1 a x'_2 \\ x_1 a^2 x_2 a x_3 &= x_1 a^2 (a^2 x_2^{-1} a^2 x_2^{-1}) x_3 = x_1 a x_2^{-1} a^2 (x_2^{-1} x_3) = x_1 a x'_2 a^2 x'_3 \end{aligned}$$

We now describe an algorithm which will apply to some arbitrary element of Pa^{-1} :

$$p = x_1 a x_2 a \cdots x_n$$

First, we apply the first or the second equation on any adjacent occurrences of a, a or a^2, a^2 to reduce it. Once we cannot do this any more, we use the first equation to "commute" occurrences of a and a^2 to put all the a s on the left. After this, we repeat until we cannot anymore. We will eventually reach something of one of the forms:

$$x_1 a x_2 \quad x_1 a^2 x_2 \quad x_1 a x_2 a^2 x_3$$

Therefore any element of P can be reduced to one of the forms:

$$x_1 a x_2 a \quad x_1 a^2 x_2 a \quad x_1 a x_2 a^2 x_3 a$$

However the third expression can be transformed

$$x_1 a (x_2 a^2 x_3 a e) = x_1 a x_2 a x'_3 a^2 x'_4 = x'_1 a^2 x'_2 a^2 x'_4 = x''_1 a x''_2$$

using the third, first, and second equation respectively. Therefore any element of P can be written in one of the forms

$$x_1 a x_2 a \quad x_1 a^2 x_2 a \quad x_1 a x_2$$

and each expression contributes at most h^2 elements.

9

- (a) Note that:

$$B(c, c) = B(B(a, b), B(a, b)) = B(a, b) = c$$

- (b) Note that:

$$B(c, x) = B((c, c), (y, x)) = B(c, (y, x)) = B((a, b), (y, x)) = B(a, x)$$

- (c) Consider the set $S = \{0, 1\} \times \{0, 1\}$ with:

$$B((a, b), (c, d)) = (a, d)$$

All the requirements can be checked quite easily.

10

The polynomial is irreducible exactly when the discriminant $b^2 - 4c$ is not a square. Adding k effectively allows us to make c arbitrary, so setting

$$c = 4^{-1}(b^2 - r)$$

the discriminant becomes r . Therefore there is a one-to-one mapping between the values of k and the resulting discriminant of the polynomial. Exactly $(p^k + 1)/2$ of the elements of $\mathbb{F}_{p^k}$ are squares, so there is our answer.

11

The statement is true. Take any derangement σ of the finite set F (that is, a permutation which does not fix any element). Then define

$$x * y = \sigma(x)$$

so that any case of associativity reduces to

$$\sigma(x) = \sigma^2(x) \iff x = \sigma(x)$$

which is never true by the derangement property.

12

Define

$$f(a) = \begin{cases} 1, & \text{if } a \in S \\ 0, & \text{if } a \notin S \end{cases}$$

and note that $f(-a) = -f(a)$. Now note that the $\langle 2 \rangle$ orbits

$$C_a = \{2^k a\} \quad k \in \mathbb{Z}, a \in F^\times$$

partition $F^\times$. We analyze the contribution to $S \cap 2S$ on each orbit. We consider each orbit as a circular graph, with an edge $a \rightarrow b$ if $b = 2a$. By assigning to each vertex $r \in \{2^k a\}$ the value $f(r)$, we obtain a cyclic binary string of length $d = \text{ord}(2)$ from each orbit. Note that the contribution to $S \cap 2S$ on each orbit is exactly the number of $1 \rightarrow 1$ edges. Let c_{ij} be the total number of edges $i \rightarrow j$. Because the total number of edges leaving and entering 1s is the same we get:

$$c_{10} + c_{11} = c_{01} + c_{11} = k \text{ (total number of edges entering/leaving 1)}$$

Note that the total number of edges leaving 0s is $d - k$ so

$$c_{00} + c_{01} = d - k$$

and substituting $c_{01} = k - c_{11}$ we obtain:

$$c_{11} - c_{00} = 2k - d \implies c_{00} + c_{11} = d \pmod{2}$$

Now we count in two cases

- Suppose that $-1 \notin \{2^k\}$. Then for every a the orbits C_a and C_{-a} are distinct and we can pair them up. Since $1 \rightarrow 1$ edges of C_{-a} corresponds exactly to $0 \rightarrow 0$ edges of C_a we see that $c_{00} + c_{11}$ is exactly the contribution to the sum from the pair $\{C_a, C_{-a}\}$, therefore every pair of orbits contributes $d \pmod{2}$. The total number of pairs of orbit is $(p^2 - 1)/2d$ so the total parity is

$$\frac{p^2 - 1}{2d} \cdot d \equiv \frac{p^2 - 1}{2} \equiv (p + 1)(p - 1)/2 \pmod{2}$$

which is even as desired.

- Now suppose that $-1 \in \{2^k\}$ and let r be minimal such that $2^r = -1$. The the number of $1 \rightarrow 1$ edges among $a \rightarrow 2a \rightarrow \dots \rightarrow 2^r a$ is the same as $0 \rightarrow 0$ edges among $2^r a \rightarrow 2^{r+1} a \rightarrow \dots \rightarrow 2^{2r} a$. Now since $f(2^r a) = f(-a) = -f(a)$, adding an edge $2^r a \rightarrow a$ does not create a $0 \rightarrow 0$ or a $1 \rightarrow 1$ edge. Therefore we form the "half loop"

$$a \xrightarrow{\quad} 2a \xrightarrow{\quad} \dots \xrightarrow{\quad} 2^r a$$

and the total number of $1 \rightarrow 1$ edges in the full loop is $c_{11}^{half} + c_{00}^{half}$. By the above argument this is $(d/2) \pmod{2}$. Since there are a total of $\frac{p^2-1}{d}$ orbits we sum to get

$$\frac{p^2-1}{d} \cdot (d/2) \equiv \frac{p^2-1}{2} \equiv (p+1)(p-1)/2 \pmod{2}$$

which is even as desired.

13

We claim that given a and b in S we can always solve the equations

$$ax = b \quad ya = b$$

at which point the set is a group (proof below). This is because we can let $x = a^{|a|-1}b$ and $y = ba^{|a|-1}$.

Now we need to show that G is a group if we can solve the equations above. Firstly there is a left identity since if we construct e_r, y so that $ae_r = a$ and $ya = b$ we see

$$ae_r = a \implies yae_r = ya \implies be_r = b$$

so since b was arbitrary e_r is a right identity. We can also construct a left identity e_l and note

$$e_l = e_l e_r = e_r \implies e_l = e_r$$

so there is an identity. Inverse follows from solving $ax = e$ and $ya = e$. The left and right inverses are the same since

$$a_l^{-1} = a_l^{-1} a a_r^{-1} = a_r^{-1}$$

as desired.

14

The claim is true. Consider the directed graph whose vertices are elements of G and whose edges correspond to multiplication by a or b . IE there is an edge $g \rightarrow h$ iff $h = ga$ or $h = gb$. We note that every vertex has exactly two edges going in and two edges going out. The problem asks us to find a Eulerian circuit. This is an application of the following theorem:

Theorem 0.1 (Eulerian Circuit Theorem). A graph has a Eulerian Circuit (a circuit which uses every edge exactly once) iff the following conditions are satisfied:

- (a) The in-degree of every vertex is equal to the out-degree.
- (b) All vertices with non-zero degree belong to a single strongly connected component.

Proof. The forward direction is clear. Therefore suppose that a graph satisfies the conditions. We will induct on the number of edges. Consider an algorithm which simply travels across any unused edge. We will eventually be stuck when we have used every "out" edge of a vertex, meaning we must have returned to the starting vertex. If the cycle includes every edge then we are done (note that by connectedness this is equivalent to using every edge of the vertices we have traveled on). Otherwise subtract the used edges and construct a Eulerian circuit on the remaining edges. We can then stitch these two circuits together by starting on a vertex which they both travel on, doing one circuit and then doing the other. $\square$

15

Set $x = y + z$ and $n = x^2$ so that we must solve:

$$n = y^2 + 1 \quad n = y^2 + 2yz + z^2 \iff z^2 + 2yz = 1$$

Note that choosing $z \in \mathbb{Z}_p^\times$ determines $y = (1 - z^2)/(2z)$ which subsequently determines $n = y^2 + 1$. We have some cases

- If the polynomial $z^2 + 2yz - 1$ is a square, it must be $(z - \sqrt{-1})^2$ which forces $y = -\sqrt{-1}$. Therefore if $p \equiv 1 \pmod{4}$ (so that $\sqrt{-1}$ exists) then there are two choices of z in $F^\times$ (namely $\pm\sqrt{-1}$) which each correspond to a unique choice for y .
- Otherwise the polynomial $z^2 + 2yz - 1$ is not a square. Therefore there are two roots which map to this y . In other words, all other choices of z have a 2-to-1 correspondance with valid values for y .

Therefore we can compute the number of ys which arise from choosing $z \in \mathbb{Z}_p^\times$

- In the case that $p \equiv 1 \pmod{4}$ the pair $\pm\sqrt{-1}$ contributes 2 distinct ys , and the other $p - 3$ choices of z contribute $\frac{p-3}{2}$ choices of y (since they pair up). Therefore there are

$$\frac{p+1}{2}$$

valid values for ys

- In the case that $p \equiv 3 \pmod{4}$, then the $p - 1$ choices of z contribute

$$\frac{p-1}{2}$$

valid values of y .

Finally, note that $y = 0$ corresponds uniquely to the degenerate solution $n = 1$. All other values of y come in pairs $\{y, -y\}$, each pair corresponding to a distinct valid value of n . Therefore the total number of reasonable ns is

- $\frac{p+3}{4}$ if $p \equiv 1 \pmod{4}$.
- $\frac{p+1}{4}$ if $p \equiv 3 \pmod{4}$.

16

Fix an element $A \in \mathcal{M}$. Note that for all $B \in \mathcal{M}$:

A, B commute or anticommute by (iii) $\implies A^2$ and B commute $\implies A^2$ commutes with everything in $\mathcal{M}$

In particular by property (iv) the square of every element of $\mathcal{M}$ is I . Now we need a second property. Find some $B \in \mathcal{M}$ which anticommutes with A by (iv) and compute

$$AB = -BA \implies A = -BAB^{-1} \implies \text{tr}(A) = 0$$

since B is invertable (by $B^2 = I$). Now I claim that the elements of $\mathcal{M}$ are linearly independent. Suppose FTSOC that there is a linear relation

$$\sum_{M \in \mathcal{M}} c_M M = 0$$

taking trace on both sides kills all the terms except I and yields:

$$\text{tr}(c_I I) = 0 \implies c_I = 0$$

Now multiply by A to get the new linear relation

$$\sum_{M \in \mathcal{M}} c_M MA = 0$$

the new term corresponding to the identity matrix is c_A since $A \in \mathcal{M}$ is the unique element which is I after multiplication by A . Therefore $c_A = 0$. Since A was arbitrary this proves linear independence, and the size of the set is bounded by n^2 since the $n \times n$ matrices have dimension n^2 .

17

We let

$$f(x) = 101x - 100 \cdot 2^x \pmod{10100}$$

By the Chinese remainder theorem we note that:

$$\mathbb{Z}/10100\mathbb{Z} \cong \mathbb{Z}/100\mathbb{Z} \times \mathbb{Z}/101\mathbb{Z}$$

The $101x$ term only contributes to the $\mathbb{Z}/100\mathbb{Z}$ term, and gives exactly the information of $a + b \pmod{100}$. The second term lives in the $\mathbb{Z}/101\mathbb{Z}$ component and gives the information of $2^a + 2^b \pmod{101}$. Therefore we only need to show that if $a + b = c + d \pmod{100}$ but $\{a, b\} \neq \{c, d\} \pmod{100}$ then $2^a + 2^b \neq 2^c + 2^d \pmod{101}$. WLOG let

$$a > b \quad c = a + k \quad d = b - k$$

and compute in $\mathbb{Z}/101\mathbb{Z}$:

$$\begin{aligned} 2^a + 2^b = 2^c + 2^d &\implies 2^a + 2^b \neq 2^{a+k} + 2^{b-k} \\ &\implies 2^{b-k}(2^k - 1) = 2^a(2^k - 1) \end{aligned}$$

There are two ways this equation can be satisfied

- If $(2^k - 1) = 0$ this implies that $k = 0 \pmod{100}$ since the multiplicative order of 2 in $\mathbb{Z}/101\mathbb{Z}$ is 100. This gives $(c, d) = (a, b)$ as desired.
- Otherwise, we can cancel $(2^k - 1)$ from both sides to get

$$2^{b-k} = 2^a \pmod{101}$$

which implies for the same reason that $k = b - a \pmod{100}$. However this means $(c, d) = (b, a)$ as desired.

18

Fix an element $a \in S$, I claim that

$$c > b \iff (a, b, c) \in S$$

induces a total order on S , where we consider a as the smallest element of S .

- By (2) $(a, y, x) \in S$ iff $(a, x, y) \notin S$ which means

$$x > y \iff y \not> x$$

which proves antisymmetry and totality.

- To prove transitivity

$$c > b, d > c \implies d > b$$

we need to show that:

$$(a, b, c), (a, c, d) \in S \implies (a, b, d) \in S$$

However condition (3) says exactly that

$$(a, b, c), (a, c, d) \in S \iff (a, b, d), (b, c, d) \in S$$

after using (2) to cycle some of the triples. This is a stronger statement so transitivity is proved.

Using the total order, we define a function $g : A \rightarrow \mathbb{R}$ such that $g(x) > g(y)$ iff $x > y$. Suppose that $g(x) < g(y) < g(z)$. Then we have that

$$(a, x, y), (a, y, z) \in S \iff (a, x, z), (x, y, z) \in S$$

by condition (3). Since we have shown that $(x, y, z) \in S$ we are done. There are some special cases I have ignored where one of the x, y, z is a , but it works out in the end.

19

Compute

$$\phi(x)\phi(e)\phi(x^{-1}) = \phi(e)\phi(x)\phi(x^{-1}) \implies \phi(x)\phi(e) = \phi(e)\phi(x)$$

so $\phi(e)$ commutes with any $\phi(x)$. Now compute

$$\phi(x)\phi(y)\phi((xy)^{-1}) = \phi(xy)\phi(e)\phi((xy)^{-1}) \implies \phi(x)\phi(y) = \phi(xy)\phi(e)$$

Now, choose $a = \phi(e)^{-1}$ and define $\psi(x) = a\phi(x)$. We can now verify the homomorphism property:

$$\psi(x)\psi(y) = (a\phi(x))(a\phi(y)) = a^2\phi(x)\phi(y)$$

Substitute our identity $\phi(x)\phi(y) = \phi(xy)\phi(e)$:

$$\psi(x)\psi(y) = a^2\phi(xy)\phi(e) = a\phi(xy)(a\phi(e))$$

Since $a = \phi(e)^{-1}$, $a\phi(e)e$ and

$$\psi(x)\psi(y) = a\phi(xy) = \psi(xy)$$

20

Let the given condition be our primary axiom:

$$(x * y) * x = y \quad \text{for all } x, y \in S$$

Substitute $x = b * a$ and $y = b$ into the axiom:

$$((b * a) * b) * (b * a) = b$$

From the original axiom, we already know that $(b * a) * b = a$. Substituting that in yields

$$a * (b * a) = b$$

as desired.

21

Consider all the element of order p . We can group them using the equivalence relation $a \sim b$ iff $a^x = b$. Therefore we have a collection of p cycles. If r is the number of p -cycles we will have $r(p-1)$ elements of order p . Therefore we need to prove that $r \equiv 1 \pmod{p}$. This reminds us of Sylow's theorems, so we attempt to prove it with a group action.

Choose a p -cycle P and let it act on the set of p -cycles. It stabilizes itself so that orbit has order 1. On the other hand it does not stabilize any other p -cycle so we have some other orbits of size p . Adding up the sizes of all the orbits gives us $r \equiv 1 \pmod{p}$ and we are done.

22

For a sequence $g_1, \dots, g_n$ let S_k denote all the products of subsequences of $g_1, \dots, g_k$. Let R_k denote the complement of S_k in G . For a choice of g_{k+1} , we want to compute the average order of R_{k+1} . Note that $|S_{k+1}| = |S_k \cup g_{k+1}S_k|$ so $|R_{k+1}| = |R_k \setminus g^{-1}R_k|$ and

$$\begin{aligned} \frac{1}{|G|} \sum_{g \in G} |R_{k+1}| &= \frac{1}{|G|} \sum_{g \in G} |R_k \setminus g^{-1}R_k| \\ &= \frac{1}{|G|} \sum_{g \in G} \sum_{h \in R_k} f_g(h) \\ &= \frac{1}{|G|} \sum_{h \in R_k} \sum_{g \in G} f_g(h) \end{aligned} \quad f_g(h) := \begin{cases} 0, & \text{if } g^{-1}h \in R_k \\ 1, & \text{otherwise} \end{cases}$$

at this point we note that $g^{-1}h$ takes every value in G exactly once. Therefore the $f_g(h) = 1$ exactly $|R_k|$ times and

$$\sum_{g \in G} f_g(h) = |R_k|$$

so subbing in we get

$$\frac{1}{|G|} \sum_{h \in R_k} R_k = \frac{|R_k|^2}{|G|}$$

in particular by pigeonhole principle for some choice of g the size of R_k decreases by at least this amount. Now define $H_k = |R_k|/|G_k|$ to be the fraction of unreachable elements and note that:

$$|R_{k+1}| \leq \frac{|R_k|^2}{|G|} \implies |H_{k+1}| \leq |H_k|^2 \implies |H_k| \leq |H_0|^{2^k}$$

We want to prove that after some logarithmly bounded number of steps we have

$$|H_n| < \epsilon \quad \epsilon := \frac{1}{|G|}$$

so that $|R_n| < 1$ forcing there to be no unreachable elements. Perform some computations

$$\begin{aligned} |H_n| &\leq |H_0|^{2^n} < \epsilon \\ 2^n \ln(|H_0|) &< \ln(\epsilon) \\ 2^n &> \frac{\ln(\epsilon)}{\ln(|H_0|)} \\ n &> \log_2 \left(\frac{\ln(\epsilon)}{\ln(|H_0|)} \right) \end{aligned}$$

therefore, since all the steps are reversible, the minimum number of steps is the ceiling of this value. Plugging in $\epsilon = 1/|G|$ and $|H_0| = 1 - \epsilon$ we see that our strictest bound is:

$$\left\lceil \log_2 \left(\frac{-\ln(|G|)}{\ln(1 - 1/|G|)} \right) \right\rceil$$

Let $x = |G|$. Playing with Taylor series gives the well known bound:

$$-\ln \left(1 - \frac{1}{x} \right) < \frac{1}{x-1}$$

Since $x \geq 2$, the term $-\ln(1 - 1/x)$ is strictly positive. Taking the reciprocal of both sides reverses the inequality:

$$\begin{aligned} \frac{1}{-\ln(1 - \frac{1}{x})} &> x-1 \\ \frac{-\ln(x)}{\ln(1 - \frac{1}{x})} &> (x-1) \ln(x) \\ \log_2 \left(\frac{-\ln(x)}{\ln(1 - \frac{1}{x})} \right) &> \log_2((x-1) \ln(x)) \\ \log_2((x-1) \ln(x)) &= \log_2(x-1) + \log_2(\ln(x)) \\ \log_2(x-1) &= \frac{\ln(x-1)}{\ln(2)} \end{aligned}$$

For $x \geq 2$, it is trivially true that $x-1 \geq \frac{x}{2}$. Applying this lower bound:

$$\frac{\ln(x-1)}{\ln(2)} \geq \frac{\ln(x/2)}{\ln(2)} = \frac{\ln(x) - \ln(2)}{\ln(2)} = \frac{1}{\ln(2)} \ln(x) - 1$$

Substituting this back into our expanded inequality, we obtain:

$$\log_2 \left(\frac{-\ln(x)}{\ln(1-1/x)} \right) > \frac{1}{\ln(2)} \ln(x) - 1 + \log_2(\ln(x))$$

As $x \rightarrow \infty$, the term $\log_2(\ln(x)) - 1$ grows and becomes strictly positive. Thus, for sufficiently large x , it can be dropped to form a strict lower bound:

$$\log_2 \left(\frac{-\ln(|G|)}{\ln(1-1/|G|)} \right) > \frac{1}{\ln(2)} \ln(|G|)$$

This demonstrates that the original quantity is bounded below by a function of the form $c \ln(|G|)$, and it holds for any positive constant $c \leq \frac{1}{\ln(2)} \approx 1.442$.

23

By the structure theorem for finite abelian groups, the structure of such a group must be of the form

$$C_{2^{k_1}} \times \cdots \times C_{2^{k_n}}$$

a product of cyclic groups of order 2^{k_i} . Now consider the following. An element of G is specified elements $(g_1, \dots, g_n)$ from each factor, and because all the orders are powers of 2

$$\text{ord}(g_1, \dots, g_n) = \max(\text{ord}(g_1), \dots, \text{ord}(g_n))$$

I know desire to compute

$$L_r = \text{number of elements of order} \leq 2^r$$

which is exactly the number of ways to choose from each factor an element of order $\leq 2^r$. Note that $C_{2^{k_i}}$ has $\min(2^{k_i}, 2^r)$ elements of order $\leq 2^r$. Therefore

$$L_r = \prod_i \min(2^{k_i}, 2^r)$$

Now note that the product of the orders of all the elements is

$$\prod_k (2^k)^{E_k} \quad E_k := L_k - L_{k-1}$$

where E_k is the number of elements order exactly 2^k . Taking $\log_2$ we want to compute

$$\begin{aligned} \sum_k k E_k &= \sum_k k (L_k - L_{k-1}) \\ &= K L_K - \sum_{j=0}^{K-1} L_j \quad K := \max(k_i) \\ &= K |G| - \sum_{j=0}^{K-1} L_j \end{aligned}$$

Now by setting this equal to 2009 and subtracting $L_0 = 1$ from both sides we get :

$$K |G| - \sum_{j=1}^{K-1} L_j = 2010$$

Define

$$d_j := \#\{k_i \text{ with } k_i \geq j\}$$

to be the number of factors of at least certain size. We notice that

$$L_r = \prod_i \min(2^{k_i}, 2^r) = \prod_{i=1}^r 2^{d_i}$$

The highest power of 2 dividing 2010 is 2^1 , forcing $c_1 = d_1 = 1$ (otherwise 4 divides all terms). Thus, G has exactly one cyclic factor, making it the cyclic group C_{2^K} . This implies $L_j = 2^j$ for all j . Substituting this into our original sum yields:

$$K2^K - \sum_{j=1}^{K-1} 2^j = 2010$$

Evaluating the geometric series and writing 2009 in its binary representation, we obtain:

$$(K-1)2^K = 11111011000_2$$

The binary representation of the right-hand side has exactly three trailing zeros, which requires $K \leq 3$. However, testing the maximum possible value $K = 3$ yields $(3-1)2^3 = 16 \neq 2008$. Therefore, no such K exists.

24

Suppose FTSOC that $a, b \in G$ are such that $a \times b \neq 0$. Then

$$a \quad a \times b \quad a \times (a \times b)$$

are three orthogonal vectors, which we will rename v_1, v_2, v_3 . We note that $-v_2 = b \times a$ is also in the group. If their magnitudes are k_1, k_2, k_3 then

$$v_1 \times v_2 = \frac{k_1 k_2}{k_3} v_3 \quad v_1 \times v_3 = -\frac{k_1 k_3}{k_2} v_2 \quad v_2 \times v_3 = \frac{k_2 k_3}{k_1} v_1$$

Now note that by applying the cross product formulas:

$$(v_1 v_2) v_2 = \frac{k_1 k_2}{k_3} v_3 \cdot v_2 = -k_1^2 v_1 \quad v_1 v_2 v_2 = v_1 (v_2 v_2)$$

This requires that $v_2 v_2$ is on the same line as v_1 since otherwise a nonzero cross product with v_1 can only be orthogonal to v_1 . Then by symmetry $v_2 v_2$ is also on the same line as v_3 and is 0. On the other hand

$$(v_1(-v_2)) v_2 = -\frac{k_1 k_2}{k_3} v_3 \cdot v_2 = k_1^2 v_1$$

also requiring $(-v_2) v_2 = 0$ by the same argument, this is impossible since $v_2 \neq -v_2$ by assumption.

25

This is actually a linear algebra problem. Take the free $\mathbb{C}$ vector space indexed by the elements of G . each g_i can be thought of as a linear transformation of the vector space via permuting the indices by the group law. Then the matrix

$$M = \frac{1}{k} \sum_{i=1}^k g_i$$

describes the evolution of the probabilities. In particular if $v_r = (v_{1r}, \dots, v_{nr})$ where v_{ar} is the probability of landing on g_a after r multiplications then $Mv_r = v_{r+1}$ and we are asked to consider:

$$\lim_{m \rightarrow \infty} \frac{1}{b^{2m}} \sum_{i=1}^n ((M^m x - v)_i)^2 \quad x = \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix} \quad v = \begin{bmatrix} 1/n \\ \vdots \\ 1/n \end{bmatrix}$$

Since the g_i form a basis, the k -steps transition probability between any two elements of G must be positive. Therefore for some k , M^k is a strictly positive matrix. IE M is primitive, so by Perron Frobenius it has a maximal eigenvalue with unique real eigenvector. Since the eigenvector v is real, it is the Frobenius eigenvector with eigenvalue 1. By the Perron projection we can write:

$$\lim_{m \rightarrow \infty} M^m = vw^T \quad w^T v = 1$$

Now note that

$$M^T = \frac{1}{k} \sum_{i=1}^k g_i^{-1}$$

which commutes with M since the group is abelian. Therefore M is a normal matrix. Since the commuting still holds in the limit

$$[vw^T, wv^T] = 0 \implies vw^T wv^T = wv^T vw^T \implies c_1 vv^T = c_2 ww^T$$

implying that w is a scalar multiple of v . We can normalize so $w = v$ and $|v| = 1$. Since a normal matrix has orthogonal eigenvectors we can write

$$M = vv^T + B \quad B \text{ diagonalizable} \quad Bvv^T = vv^T B = 0$$

Now write:

$$\sum_{i=1}^n ((M^m x - v)_i)^2 = \sum_{i=1}^n ((vv^T + B^m)x - v)_i^2 = \sum_{i=1}^n (B^m x)_i^2$$

at which point we can write x in terms of the eigenvectors of B and set b to be the largest magnitude of one of the eigenvalues.

26

I claim that there is a right identity of S . For some $a \in S$ find $e_r \in S$ with

$$ae_r = a$$

then find y with $ya = b$ and left multiply so

$$be_r = b$$

and since b is arbitrary e_r is a right identity. If we find d such that $cd = e_r$ we see that

$$ac = bc \implies acd = bcd \implies ae_r = be_r \implies a = b$$

as desired.

27

Consider the subgroup N formed by expressions of this form (without the length condition). We first must show $g, h \in N$ so that $N = G$. Note that

$$g(hg^{-1})^{|G|-1}h = ggh^{-1}h = g^2$$

is in N . Therefore since g has odd order $g \in N$ and now $h \in N$ as well.

Now we must address the length condition. Suppose we have an expression of length larger than $|G|$:

$$g^{m_1} h^{n_1} g^{m_2} h^{n_2} \dots$$

and write l_k to be the product of the first k terms. By pigeonhole $l_k = l_r$ for some k, r . Then cutting out all the terms in between l_k and l_r does not change the expression while reducing the length.

We let $[x] = x - \lfloor x \rfloor$ denote the fractional part of x . Consider

$$\begin{aligned} a_k &= \left\lfloor \frac{mk}{n} \right\rfloor - \left\lfloor \frac{m(k-1)}{n} \right\rfloor \\ &= \begin{cases} \left\lceil \frac{m}{n} \right\rceil, & \left\lfloor \frac{m}{n} \right\rfloor + \left\lceil \frac{m(k-1)}{n} \right\rceil \geq 1 \\ \left\lfloor \frac{m}{n} \right\rfloor, & \text{otherwise} \end{cases} \\ &= \begin{cases} \left\lceil \frac{m}{n} \right\rceil, & (k-1)m \in \{-1, \dots, -m\} \pmod{n} \\ \left\lfloor \frac{m}{n} \right\rfloor, & \text{otherwise} \end{cases} \end{aligned}$$

Since $\gcd(m, n) = 1$ we can find a unique x such that:

$$xm \equiv 1 \pmod{n}$$

Now consider

$$\begin{aligned} a_{k+x} &= \begin{cases} \left\lceil \frac{m}{n} \right\rceil, & (k-1+x)m \in \{-1, \dots, -m\} \pmod{n} \\ \left\lfloor \frac{m}{n} \right\rfloor, & \text{otherwise} \end{cases} \\ &= \begin{cases} \left\lceil \frac{m}{n} \right\rceil, & (k-1)m \in \{0, \dots, -m+1\} \pmod{n} \\ \left\lfloor \frac{m}{n} \right\rfloor, & \text{otherwise} \end{cases} \\ &= \begin{cases} a_1, & k = n \\ a_n, & k = 1 \\ a_k, & \text{otherwise} \end{cases} \end{aligned}$$

Let

$$W = gh^{a_1}gh^{a_2} \dots gh^{a_n} = e$$

and note by conjugating the equation that a cyclic shift of W must also evaluate to e , so shifting indices by x yields:

$$gh^{a_{1+x}}gh^{a_{2+x}} \dots gh^{a_{n+x}} = e$$

Since $a_{k+x} = a_k$ for all $k \notin \{1, n\}$, we equate the words and cancel all identical terms to isolate the shifted indices at $k = n$ and $k = 1$:

$$\begin{aligned} gh^{a_n}gh^{a_1} &= gh^{a_{n+x}}gh^{a_{1+x}} \\ gh^{a_n}gh^{a_1} &= gh^{a_1}gh^{a_n} \end{aligned}$$

Substitute $a_n = \lceil m/n \rceil$ and $a_1 = \lfloor m/n \rfloor$:

$$\begin{aligned} gh^{\lceil m/n \rceil}gh^{\lfloor m/n \rfloor} &= gh^{\lfloor m/n \rfloor}gh^{\lceil m/n \rceil} \\ h^{\lceil m/n \rceil}gh^{\lfloor m/n \rfloor} &= h^{\lfloor m/n \rfloor}gh^{\lceil m/n \rceil} \end{aligned}$$

Cancel $h^{\lfloor m/n \rfloor}$ from both sides:

$$hg = gh$$

as desired.

We can write power series

$$\frac{1}{1-xy} = 1 + xy + xyxy + \dots \quad \frac{1}{1-yx} = 1 + yx + yxyx + \dots$$

which is unrigorous but motivates the guess

$$(1-yx)^{-1} = 1 + y(1-xy)^{-1}x$$

which can be verified using direct algebraic manipulation.

30

By our solution to problem 29 we have

$$(1 - yx)^{-1} = 1 + y(1 - xy)^{-1}x$$

and subbing this into the equation yields the desired identity.

31

By the fundamental theorem of finite abelian groups, G is isomorphic to the direct product of its Sylow p -subgroups. Assume, for the sake of contradiction, that a Sylow p -subgroup P of G is not cyclic. Consequently, the prime-power decomposition of P must contain at least two cyclic factors. This implies P (and therefore G) contains a subgroup isomorphic to $\mathbb{Z}_p \times \mathbb{Z}_p$. The elementary abelian group $\mathbb{Z}_p \times \mathbb{Z}_p$ has multiple distinct subgroups of order p (take $\langle 1, 0 \rangle$ and $\langle 0, 1 \rangle$). This contradicts the hypothesis that G possesses exactly one subgroup of order p . Thus, every Sylow p -subgroup of G must be cyclic. Since G is the direct product of cyclic groups with pairwise coprime prime-power orders, G is cyclic by Chinese remainder theorem.

32

I first claim that f almost preserves the order of real numbers. That is:

$$a \geq b \implies f(a) \geq f(b)$$

Let $a = b + r$ for $r \in \mathbb{R}^+$. Since

$$f(a) - f(b) = f(a - b) = f(r)$$

we only need to show that $f(r) \geq 0$. Since r is a positive real number it is x^2 for some x . Now

$$f(r) = f(x^2) = f(x)^2 \geq 0$$

as desired.

Now I claim that f fixes the rational numbers. Note

$$1 = f(1) = f(\underbrace{1/n + \dots + 1/n}_{n \text{ times}}) = nf(1/n) \implies f(1/n) = 1/n$$

and note $f(a/b) = af(1/b) = a/b$ as desired.

Finally I claim that f fixes $\mathbb{R}$. Suppose for the sake of contradiction that $f(x) \neq x$. Then we can find some $\epsilon > 0$ with $f(x) \notin B_\epsilon(x)$ (ball of radius epsilon around x). By density of rational numbers we find rational $r_1, r_2 \in B_\epsilon(x)$ with $r_1 < x < r_2$. Then by order preserving property

$$r_1 \leq f(x) \leq r_2 \implies f(x) \in B_\epsilon(x)$$

a contradiction.

33

(a) Note that

$$[a, b] = aba^{-1}b^{-1} = abab = 1$$

so we have a finite $\dim \mathbb{F}_2$ vector space, hence finite.

(b) Note the fundamental relation $abab = b^2a^2$ which implies

$$bab = a^2b^2a^2 \quad b^2ab = ba^2b^2a^2$$

where the first implies the second. It suffices to show the free group

$$G = \langle x_1, \dots, x_n : g^3 = 1 \rangle$$

is finite. We will induct on the number of generators to show that every element of can be expressed in a bounded number of terms, at which point it follows that G is finite. Consider an element $g \in G$ and keep track only of occurrences of x_1 or x_1^2 .

$$g = -x_1 - x_1^2 - x_1^2 - \dots$$

where each $-$ denotes some expression of $x_2, \dots, x_n$. We can use the second relation to "commute" adjacent occurrences of x_1 and x_1^2

$$-x_1^2 - x_1 - \mapsto -x_1 - x_1^2 -$$

Therefore we can gather all the x_1 on one side and all the x_1^2 on the other.

$$g = -x_1 - \dots - x_1 - x_1^2 - \dots - x_1^2 -$$

at which point the first relation is used to reduce adjacent occurrences of x_1, x_1 and x_2, x_2

$$-x_1^2 - x_1^2 - \mapsto -x_1 - \quad -x_1 - x_1 - \mapsto -x_1^2 -$$

reducing the total number of occurrences. We can repeat these steps until we cannot continue, reducing to an expression of one of the forms

$$- \quad -x_1 - \quad -x_1^2 - \quad -x_1 - x_1^2 -$$

where in every case the minimum number of terms is bounded by $A_n \leq 2 + 3A_{n-1}$ where A_{n-1} is the bound on $n - 1$ generators by the inductive hypothesis.

34

The action of G on itself by right multiplication induces an embedding

$$G \rightarrow S_{|G|}$$

into the permutations on the elements of G . Since 2 divides $|G|$, we may find an element $x \in G$ which has order 2. The action in $S_{|G|}$ therefore looks like $2n + 1$ disjoint 2 cycles, so the sign of this permutation is $(-1)^{2n+1} = -1$. In particular the composition

$$G \rightarrow S_{|G|} \xrightarrow{\text{sgn}} \{\pm 1\}$$

has full image. Therefore the kernel is an index 2 normal subgroup.

35

I first claim that if there is a two sided inverse, it is the only right inverse. Let a be a two sided inverse to x and b a right inverse to x . Then

$$a = axb = b$$

proving that there cannot be a two sided inverse. Let y be a right inverse to x and let $w = 1 - yx$ which is nonzero. Note that

$$xw = x - xyx = x - x = 0$$

so the elements

$$y_n = y + wx^n$$

are all right inverses to x . I claim that they are distinct. Suppose FTSOC that

$$\begin{aligned} y_{n_1} = y_{n_2} &\implies wx^{n_1} = wx^{n_2} \\ &\implies wx^k = w && \text{right multiply by } y^{n_2} \text{ and define } k := n_1 - n_2 \\ &\implies wx^{k-1} = wy && \text{right multiply by } y \end{aligned}$$

at this point we note that

$$wy = y - yxy = y - y = 0$$

so we have

$$wx^{k-1} = 0 \implies w = 0$$

after right multiplying by y^{k-1} . This is a contradiction as desired.

36

It suffices to write $xyz + xyz$ as a sum of squares. Note that

$$(x + yz)^2 = xyz + yzx \quad (xy + z)^2 = xyz + zxy \quad (y + zx)^2 = yzx + zxy$$

Therefore:

$$(x + yz)^2 + (xy + z)^2 - (y + zx)^2 = 2xyz$$

37

Note that

$$(a + b)(y + x) = (a + b)y + (a + b)x = ay + by + ax + bx$$

and

$$(a + b)(x + y) = a(y + x) + b(y + x) = ay + ax + by + bx$$

so setting these equal

$$ay + by + ax + bx = ay + ax + by + bx$$

and cancelling the leftmost and rightmost terms from both sides yields

$$by + ax = ax + by$$

as desired.

38

To solve this problem I will sequentially compute the following values:

$C_n(m)$: Let $C_n(m)$ be the number of way to pick m linearly independant vectors in $\mathbb{F}_p^n$ (order mattering). If we pick the vector sequentially, then the k th vectors can be chosen to be anything not in the span of the previous $k - 1$ vectors. By linear independance there are p^{k-1} elements in this span, therefore we have:

$$C_n(m) = (p^n - 1)(p^n - p)(p^n - p^2) \cdots (p^n - p^{m-1}) = \prod_{k=0}^{m-1} (p^n - p^k)$$

I_n : Let I_n be the number of invertable $n \times n$ matrices. This is just the number of ways to pick n linearly independant columns for the matrix. Therefore:

$$I_n = C_n(n) = \prod_{k=0}^{n-1} (p^n - p^k)$$

$S_n(d)$: Let $S_n(d)$ be the number of dimension d subspaces of $\mathbb{F}_p^n$. First, we choose d linearly independent vectors spanning the subspace, and there are $C_n(d)$ way to do this. On the other hand, there are multiple choices of spanning vectors that generate the same subspace V . To account for this, note that any two basis sets for V are related uniquely by a $\dim V \times \dim V$ change of basis matrix. Therefore divide out by I_d to obtain:

$$S_n(d) = \frac{C_n(d)}{I_d} = \frac{\prod_{k=0}^{d-1} (p^n - p^k)}{\prod_{k=0}^{d-1} (p^d - p^k)}$$

$R_n(r)$: Let $R_n(r)$ be the number of $n \times n$ matrices of rank r . To pick a matrix M in this set, we need to choose the kernel space, and there are $S_n(n-r)$ ways to do this. If we extend a basis of $\ker M$ to a basis of $\mathbb{F}_p^n$ via $v_1, \dots, v_r$ then M is uniquely determined by the image of the v_i . However the images must be linearly independent so that the rank of M is indeed r . Therefore there are $C_n(r)$ choices for the images. This gives:

$$R_n(r) = S_n(n-r) \cdot C_n(r) = \frac{\prod_{k=0}^{n-r-1} (p^n - p^k)}{\prod_{k=0}^{n-r-1} (p^{n-r} - p^k)} \prod_{k=0}^{r-1} (p^n - p^k)$$

39

(a) Consider an element $c = (c_1, c_2, \dots)$. Let

$$c_i^{(k)} = \begin{cases} 0 & \text{if } k \leq i \\ c_i & \text{otherwise} \end{cases}$$

be the same sequence with the first few terms set to 0. Note that $f(c - c^{(k)}) = 0$ since it is a finite sum, therefore

$$f(c) = f(c - c^{(k)} + c^{(k)}) = f(c - c^{(k)}) + f(c^{(k)}) = f(c^{(k)})$$

so f does not care about any finite amount of initial terms.

We call a sequence $d = (d_1, d_2, \dots)$ *p-increasing* if p^k divides d_k for all k . Since every term of $d^{(k)}$ is divisible by p^k we can write

$$f(d) = f(d^{(k)}) = p^k f(d^{(k)}/p^k)$$

implying that $f(d)$ is divisible by p^k . Since k was arbitrary, this is only possible if $f(d) = 0$, so f kills every *p-increasing* sequence.

Now we want to show that $f(c) = 0$. By bezouts theorem we can find x_i, y_i for every i such that

$$2^i x_i + 3^i y_i = c_i$$

so define the sequences $a = (2x_1, 2^2x_2, \dots)$ and $b = (3y_1, 3^2y_2, \dots)$. By construction we have

$$f(c) = f(a + b) = f(a) + f(b)$$

however a, b are 2-increasing and 3-increasing respectively, therefore they are killed by f . This gives $f(c) = 0$ as desired.

(b) Suppose for the sake of contradiction that there are infinitely many indices $i_1 < i_2 < \dots$ such that $f(e_{i_k}) \neq 0$. We will construct a sequence $c = (c_1, c_2, \dots)$ which yields a contradiction via the binary representation of $f(c)$. Define $c_{i_k} = 2^{m_k}$ and $c_j = 0$ for $j \notin \{i_1, i_2, \dots\}$. We choose the exponents m_k recursively to grow fast enough to prevent binary overlap:

$$m_1 = 1, \quad m_{k+1} = m_k + \lceil \log_2 |f(e_{i_k})| \rceil + 2$$

Since $c \in G$, its evaluation $f(c)$ must equal some finite integer N . For any n , we can split c into a finite prefix and an infinite tail:

$$c = \sum_{k=1}^n 2^{m_k} e_{i_k} + 2^{m_{n+1}} d^{(n)}$$

where $d^{(n)}$ is an integer sequence (since all subsequent terms have at least a factor of $2^{m_{n+1}}$). Applying f to both sides, we get:

$$N = f(c) = \sum_{k=1}^n 2^{m_k} f(e_{i_k}) + 2^{m_{n+1}} f(d^{(n)})$$

This implies the congruence:

$$N \equiv \sum_{k=1}^n 2^{m_k} f(e_{i_k}) \pmod{2^{m_{n+1}}}$$

Notice our choice of m_{k+1} ensures that the magnitude of $2^{m_k} f(e_{i_k})$ is strictly less than $2^{m_{k+1}-1}$. Because the blocks of bits for each term are separated by zeros, each $f(e_{i_k}) \neq 0$ forces at least one non-zero bit into the 2-adic representation of N between the positions m_k and m_{k+1} .

Since there are infinitely many such k , the binary representation of N must contain infinitely many non-zero bits (or infinitely many alternating bit flips if $N < 0$). However, N is a fixed, finite integer, meaning its binary representation must eventually become all 0s (if $N \geq 0$) or all 1s (if $N < 0$). This is a contradiction. Therefore, $f(e_i) = 0$ for all but finitely many i .

40

We must compute the sum:

$$\sum_{g \in G} \sum_{\substack{x \in G \\ xg=gx}} 1 = \sum_{g \in G} C_G(g)$$

Now note that the map

$$G \times G \rightarrow G \quad u \times v \mapsto uvu^{-1}$$

by conjugation defines a G action on itself, and we are tasked with finding the sum

$$\sum_{g \in G} |X^g|$$

where X^g are the fixed points of G under conjugation by g . This is because commuting is equivalent to being fixed by the conjugation action. Then by Burnside's lemma this is equal to

$$|G| |X/G|$$

where X/G is the number of orbits under this action. IE the number of conjugacy classes. The answer is therefore:

$$|G| k(G)$$

41

This turned out to be a difficult problem about S_n reps. We work in the group algebra $\mathbb{C}[G]$. Let

$$K = \sum_{c \in \{n\text{-cycles}\}} c$$

be the conjugacy class sum of the n -cycles. We need to compute the coefficient in front of the identity in K^3 . This is exactly the trace in the regular representation, so we use Fourier analysis and sum over the characters:

$$\text{Ans} = \sum_{V \in \text{Irr } S_n} d_V \text{tr } K^3|_V \quad d_V = \dim V$$

We note that the sum over a conjugacy class C is scaling by $|C| \chi_V(C)/d_V$ via standard rep theory. Therefore the trace of K^3 is $\frac{|C|^3 \chi_V(C)^3}{d_V^2}$. Therefore we must calculate:

$$\sum_{V \in \text{Irr } S_n} \frac{|C|^3 \chi_V(C)^3}{d_V^2}$$

The difficulty now is computing the character.

